1.6b

Sensation
vision

How do we see light?

Our eyes turn light energy into neural messages
Our brain then creates what we consciously see

Stimulus Energy: light energy

Visible light is is a thin slice of the spectrum of electromagnetic energy.

This spectrum ranges from 10^{-5} nm Gamma Rays to 400-700 nm visual light to 10^{13} nm Broadcast Bands.

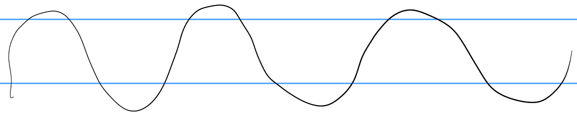
Light travels in waves.

Wavelength: distance from one peak to next

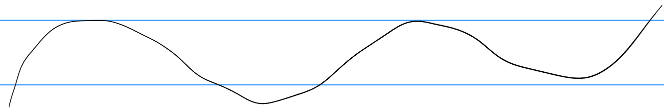
Wave length determines hue

amplitude: height determines intensity: # of energy in wave

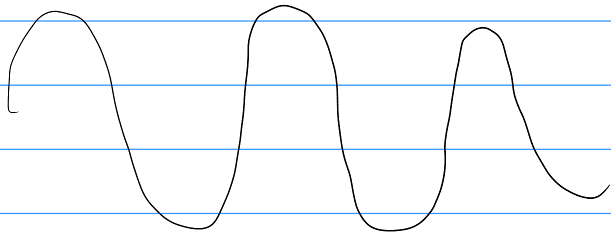
intensity influences brightness



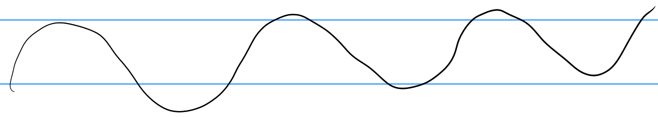
Shorter wavelength
high frequency
bluish colors



longer Wavelength
low frequency
reddish colors



Great amplitude
bright colors



Small amplitude
dull colors.

The eye

light enters the eye through the cornea, which bends light to help focus

Then light passes through the pupil, a Small, adjustable opening.

Iris controls the size of the pupil. Colored muscle dilates or constricts in response to light intensity.

Each Iris is distinct.

Iris responds to cognitive & emotional state.

Imagine sunny sky → Iris constricts
Imagine dark room → Iris dilates

When you are disgusted, or about to answer no → Iris constricts
When you feel amorous or trusting, → Iris dilates

After passing thru pupil, light hits lens
Transperant lens focuses rays into an image on retina ::
multilayered tissue lining the back inner surface of eyeball.
to focus rays, lens changes its curvature and thickness :: accommodation

Myopia (nearsightedness) :: lens focuses image on a point in front of the retina.
U can see near objects clearly, not far ones.
Can be remedied with glasses, contact lenses or surgery.

Farsightedness :: seeing distant objects better than near ones, Occurs when lens focuses on point behind retina.

Image passing through lens is cast upside down on the Retina.

Personal Connection:
I am nearsighted.

Retina's millions of receptor cells convert particles of light energy into neural impulses, & foward them to the brain.
Brain reassembles them, rightside up creating the image we perceive

A baseball batter takes $4/10$ sec to respond to a fastball. Visual information travels at abstract levels in order to acheive this speed.

How does info go from our eye to our brain

After making way thru retina's outer layer, and reaching the very back, a particle of light would encounter the Retina's Photoreceptor Cells.

Rods :: retinal receptors that detect black, white & gray, and are sensitive to movement. Necessary for peripheral & twilight vision, when cones don't respond.

Cones :: retinal receptors that are concentrated near center of Retina.
Function in well-lit conditions.
Detect fine details, color

At photoreceptor cells, the light energy triggers chemical changes.
Chemical reaction sparks neural signals in nearby bipolar cells.
The bipolar cells would then activate neighbouring ganglion cells, whose axons twine together like a rope, forming an optic nerve.

After a momentary stop at the thalamus, the info would go to the visual cortex.

The optic nerve is very fast. Can send almost 1 million concurrent messages, through its almost 1 million ganglion fibers.

Where the optic nerve leaves the eye, the eye has a blind spot, with no receptor cells.
Brain automatically fills this hole in.

Cones cluster in & around the fovea: the retina's area of central focus.
Here, cones typically have their own bipolar cell, which connects to a dedicated ganglion cell, which relays info to the visual cortex. (Where a large area is dedicated to the fovea)
Cones detect black & white & enable color perception, but not in the dark.
You have 6 million cones.

Rods congregate in the Retina's outer regions.

They remain sensitive in dim vision.

They have no hotline to the brain

They enable black & white vision

Several cones pool their faint energy output into a single bipolar cell.

120 million rods

In darkness, pupils adapt to allow more light to reach retina.

Fully adapting takes 20 min.

At entry level, retina's nu layers also help encode & analyze info.

ex. In bug eyes, 3rd nu layer contains bug detector cells

In our eyes, any retinal area relays its info to a corresponding area in visual cortex

Half of each eye's FOV is sent to each half of the brain.

Each optic nerve splits in 2 at the optic chasm

Retinal cells are so sensitive, even pressure can trigger them.

How does color emerge?

Color is an invention of our mind

"The [light] rays are not colored" - Newton, 1704

A tomato can be thought of as any color but red, as if reflects red light but absorbs all others.

In 19th century, Hermann von Helmholtz built on insight of Thomas Young:

∴ all colors can be made by red, green & blue

∴ eye has 3 kinds of color receptors

This was the Young-Helmholtz trichromatic theory

Our Retinas have 3 types of receptors, each especially sensitive to red, green or blue light.

When light stimulates combinations of these cones, we see other colors.

$1/12$ male $1/200$ female have genetically sex-linked condition of color-deficient vision.

They are not 'color blind'. They lack functioning red-sensitive and/or green-sensitive cones

Their vision can be monochromatic or dichromatic instead of trichromatic

But people blind to red & green often see yellow.

To answer this, Ewald Hering looked at afterimages

Staring at a green shape for a long time, then a white sheet of paper, then you'll see red green's opponent color:

From this, Herring hypothesized 2 more processes. One for red-green inversion, & one for blue-yellow inversion.

This hypothesis became opponent-process theory

Color vision depends on 3 sets of opposing retinal processes:

Red - Green
Blue - Yellow
White - Black

As impulses travel to visual cortex, some neurons in the retina & thalamus are turned 'on' by red, but 'off' by green. Others are opposite.

"red" and "green" messages cannot travel at once
∴ we see either red or green, not a mixture

red & blue travel in seperate channels
∴ we can see a combination, magenta.

Color Processing occurs in 2 stages

1) With Y-H TC theory, retina's red, green & blue sensitive cones respond in varying degrees

2) The cone's responses are then processed by opponent-process cells

Feature Detection

Our brain de constructs visual images & re constructs them

Some neurons fired when shown lines at one angle, different neurons fired. These feature detectors receive info from individual ganglion cells.

They pass this to other areas, where teams called super cell clusters respond to more complex patterns.

For important objects & events, we have a 'vast visual encyclopedia', in the form of specialized cells.

These cells respond to one type of frequent stimuli.

One region of temporal lobe is essential for facial - recognition.

Parallel Processing

The brain processes various subdimensions, motion, form, depth, color, etc. simultaneously

For facial recognition, brain compares retinal info to stored info.

It looks more likely this info is distributed over a network of cells rather than just 1.

But some 'super cells' do appear to respond selectively to 1 or 2 faces out of 100.

Strokes can damage these areas, leading to face blindness, prosopagnosia, or blindsight:
ex. shown sticks they report seeing nothing. When asked the direction, they are correctly able to answer horizontal or vertical.

MCQ: 1c, 2b, 3b, 4b, 5c